

ADITYA SURENDRAN

Technology & Computer Science Student | Software • AI • Distributed Systems

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PROFILE

Technology-focused Transition Year student with a strong interest in software engineering, artificial intelligence, distributed systems and emerging technology. I independently research, design and develop technical projects using AI-assisted development workflows to prototype, test and iterate on complex ideas. My work has earned multiple first-place SciFest awards, a CREST Gold Award, and selection for international science and AI opportunities. I am seeking hands-on work experience in a professional technology environment where I can learn from experienced developers and contribute to real projects.

EDUCATION

Athlone Community College

Transition Year • Athlone, Co. Westmeath • 2026

- Strong interests in Computer Science, Mathematics, Science and Technology; planning to study Computer Science at university.
- Student Council Executive / Junior Officer.

PROJECTS & RESEARCH

SUBLYNE / BYZANTINE FAULT TOLERANCE

Founder & Developer • Independent Distributed Systems Research

- Research into making Byzantine Fault Tolerant consensus more scalable while maintaining security guarantees.
- Investigated Byzantine faults, consensus, cryptographic signatures, BLS signature aggregation and communication complexity.
- Designed, developed and evaluated experimental consensus systems and simulations using throughput, latency, communication complexity and fault tolerance.
- Led the project from research and system design through experimentation, iteration and presentation using AI-assisted development workflows.
- Recognition: SciFest@School Best Project / 1st Place; SciFest@College TUS Athlone Best Project and 1st Place – Intermediate Technology; progressed to SciFest National Final 2026.

SMARTER CANCER STAGING

Independent Machine Learning Research • June 2025 – January 2026

- Investigated whether gene-expression data alone could provide an early signal associated with broad breast cancer stage categories.
- Analysed TCGA and independent international RNA-seq data, using feature selection and compact 30–50 gene panels.
- Compared Logistic Regression, Linear SVM, Random Forest and XGBoost using cross-validation, held-out testing and external validation.
- Built an ESP32-S3 edge device for local preprocessing and model inference with an integrated display and 3D-printed enclosure.
- Recognition: British Science Association CREST Gold Award.

PARETO-OPTIMAL URBAN TRANSPORT

Environmental & Transport Research

- Investigated urban transport optimisation while incorporating environmental and health externalities.
- Recognition: ECO-UNESCO Best Junior Transport Project; EirGrid Cleaner Climate Award.

RESEARCH & DEVELOPMENT APPROACH

- Comfortable independently researching unfamiliar technical subjects and turning research questions into experiments and prototypes.
- Use AI-assisted development tools for prototyping, implementation, debugging and iteration while directing project architecture, requirements, evaluation and technical decisions.
- Experienced communicating complex technical ideas through reports, demonstrations, presentations and competition judging.

ACHIEVEMENTS & EXPERIENCE

International recognition, competition results, leadership and interests

INTERNATIONAL RECOGNITION

Intel AI Global Summit

Selected to represent Ireland in an international AI and technology opportunity connected with the Intel AI Global Summit.

Eskom Expo International Science Fair

Selected to represent SciFest and Ireland at the Eskom Expo International Science Fair in South Africa, 29 September – 2 October 2026.

SCIENCE & TECHNOLOGY AWARDS

- SciFest@College TUS Athlone 2026: Best Project and 1st Place – Intermediate Technology.
- SciFest@School 2026: Best Project / 1st Place.
- ECO-UNESCO Young Environmentalist Awards 2026: Best Junior Transport Project.
- EirGrid Cleaner Climate Award: awarded for the urban transport research project at SciFest@School 2026.
- British Science Association CREST Gold Award: awarded for the Smarter Cancer Staging research project.

MATHEMATICS & PROBLEM SOLVING

LWETB Junior Maths Competition 2025

1st Place across participating Longford and Westmeath ETB schools, qualifying for the ETB All-Ireland Finals.

ETB All-Ireland Junior Maths Competition 2025

2nd Place nationally among ETB schools; the team remained tied for first for much of the competition.

ETB All-Ireland Junior Maths Competition 2024

4th Place / Finalist nationally among ETB schools.

IAMTA Junior Regional Problem Solving Competition 2026

1st Place as part of a team in the regional junior problem-solving competition.

IMTA Maggie Gough Outstanding Achievement 2024

5th overall across all participating year groups and 2nd among Second Year students; scored 95% in a competition involving more than 15,000 students from over 400 schools.

LEADERSHIP

Student Council Executive • Athlone Community College

- Student Council Officer / Junior Officer.
- Work with students and staff on school initiatives and represent student interests.
- Develop communication, collaboration and organisational skills alongside academic and technical work.

TECHNICAL INTERESTS

Artificial Intelligence • AI-Assisted Development • Software Engineering • Distributed Systems • Machine Learning • Blockchain • Cryptography • Embedded Systems • Edge Computing • Virtual Reality • Robotics • Emerging Technology

STRENGTHS

- Independent learning: regularly learn unfamiliar technologies and research areas independently.
- Problem solving: strong competitive mathematics and technical research background.
- Research: experience designing experiments, evaluating evidence and identifying limitations rather than only optimising headline results.
- Communication: experienced presenting technical projects to judges, educators and professional audiences.
- Curiosity: motivated by understanding how complex systems work and building experimental solutions.

ONLINE

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References available on request.